Aditya Tiwari

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Experience

JTP (Japan Third Party Co., Ltd.) Mar 2026 – Nov 2026
System Engineer Trainee *Remote from India*

- Worked as a Systems Engineer on JTP's internal robotics automation projects, testing behaviour scripts for SoftBank **NAO humanoid robots** and diagnosing execution failures from runtime logs.
- Built a **Python log-parsing tool** that scanned test-run logs to isolate failing behaviours and route error details to the owning developers, cutting manual log review.
- Resolved **client-reported bugs and software issues**, reproducing each problem, identifying the root cause and delivering fixes, with written project reports to the Japan-based team.
- Delivered a containerized content recommendation platform (React, FastAPI, PostgreSQL; **3 Docker Compose services**, **5 REST endpoints**) over a ∼10,000-title catalog, with per-query TF-IDF ranking that avoids an ∼**800 MB** precomputed matrix.
- **Tech Stack:** Python, NAO robotics, Machine Learning, Pandas, Scikit-Learn, SQL, FastAPI, React, PostgreSQL, Docker, Jenkins, REST APIs, Data Validation, Git

PwC (Salesforce Launchpad Program) Jan 2025 – Jul 2025
Software Engineering Trainee *Remote, India*

- Shipped **3 enterprise-grade applications** in a 6-month SWE program using OOP design patterns, modular code and database optimization across 4 certified tracks (Java, SQL, RDBMS, Salesforce CRM).
- Designed normalized SQL schemas across 15+ tables and tuned queries with indexing, cutting avg. query time by ∼**35%**.
- Delivered features across 6 Agile/Scrum sprints with peer code reviews, and configured 20+ Salesforce CRM automation rules.

Altair Apr 2025 – Jun 2025
Data Science Intern · Grade: Excellent *Remote, India*

- Built ML classification and clustering pipelines in RapidMiner over 100K+ records, using cross-validation and hyperparameter tuning to reach **92% model accuracy**.
- Automated preprocessing (correlation, PCA, outlier removal), cutting feature dimensionality by ∼30% and experiment cycles by ∼25%.
- **Tech Stack:** Altair RapidMiner Studio, RapidMiner Server, Process Pipelines, CRISP-DM, Python, R, SQL, Machine Learning

Technical Skills

Languages: Python, Java, JavaScript, TypeScript, SQL, HTML5, CSS3
Web & Backend: Spring Boot, Spring Data JPA, Hibernate, Spring Security, FastAPI, React JS, REST APIs, Kafka, Microservices
Databases: MySQL, PostgreSQL, MongoDB, Redis, SQLAlchemy, Flyway, CTEs, Window Functions, Query Optimization
Testing & DevOps: JUnit 5, Mockito, Selenium, Playwright, pytest, Locust, Regression Testing, CI/CD (GitHub Actions, Jenkins)
Cloud & Tools: AWS (EC2, S3, RDS), Docker, Kubernetes, Prometheus, Grafana, Splunk, OpenTelemetry, Linux, Bash, Git, GitHub
AI & ML: PyTorch, TensorFlow, Scikit-Learn, OpenCV, NLP, Hugging Face, LangChain, OpenAI API
Core CS: Data Structures & Algorithms, OOP, Design Patterns, Multithreading, System Design, DBMS, OS, Computer Networks
Practices: SDLC, Agile/Scrum, Kanban, Code Reviews, Secure Coding, OWASP, PDCA, Troubleshooting

Education

GITAM University, Visakhapatnam 2022 – 2026
B.Tech in Computer Science Engineering (Graduated 2026) *CGPA: 7.44*

Relevant Coursework: Data Structures, DBMS, Operating Systems, Computer Networks, Software Engineering, Machine Learning

Projects

ServScope | *Distributed Tracing & Observability Platform* 2026 · GitHub

- Built an observability platform that ingests OpenTelemetry traces and logs from a 4-service app, builds per-minute metrics at ingest, and links alerts to the traces showing the root cause, with a React + TypeScript dashboard.
- Designed reliable, idempotent ingestion (data confirmed only after commit, duplicate-safe keys, 503 backpressure), verified by failure drills (database outage, crash mid-write, traffic flood).
- Profiled with py-spy and rewrote the hot database path in SQLAlchemy Core, raising throughput **about 5x (3,400 to 17,000 spans/s** on one core), measured with Locust; CI runs **220 tests**, Playwright and security scans.
- **Tech stack:** Python, FastAPI, PostgreSQL, SQLAlchemy, OpenTelemetry, React, TypeScript, Docker, Locust, GitHub Actions

ClaimFlow | *Event-Driven Health Claims Processing* 2026 · GitHub

- Built two Spring Boot microservices that process health-insurance claims over Apache Kafka, checking coverage, covered services, deductible, coinsurance and annual maximum to approve, deny or pend each claim.
- Published events only after the database commit, made decision handling idempotent, and added retries with a dead-letter topic; implemented polyglot persistence with MySQL (Flyway migrations) for claims, members and plans, and MongoDB for an append-only audit timeline.
- Covered by **32 JUnit 5 tests** (embedded Kafka, full two-service flow) and **4 Selenium UI tests**; GitHub Actions runs them plus a Docker Compose smoke test, with structured logs shipped to Splunk.
- **Tech stack:** Java, Spring Boot, Apache Kafka, MySQL, MongoDB, JUnit 5, Selenium, Docker, GitHub Actions, Splunk

AI Med Card | *Full-Stack Healthcare Platform* 2026 · GitHub · Live Demo

- Built a healthcare platform with Patient, Doctor and Superadmin portals on a FastAPI REST backend and PostgreSQL, with bcrypt + JWT role-based auth, deployed live with Docker on Render.
- Built provider-agnostic AI report analysis (Claude or OpenAI): PDF text extracted with pypdf and trimmed to fit model context limits, photographed reports sent to vision models, with per-user rate limiting; covered by **19 pytest API tests** and Playwright end-to-end tests.
- **Tech stack:** Python, FastAPI, PostgreSQL, SQLAlchemy, JWT, pytest, Playwright, Docker, Render, JavaScript, HTML5, CSS3

Certifications & Achievements

- **Certifications:** PwC Salesforce Launchpad (Java, RDBMS, Salesforce CRM); NPTEL Data Science (IIT); Coursera: Python Data Structures, C for Everyone, Cloud Computing (IBM); GenAI Workshop, IIT Hyderabad
- Top teams nationally at **Walmart Global Tech Hackathon 2025** (500+ teams); built and demoed KrishiBal at **Smart India Hackathon 2025** (live).
- E-Club Co-Coordinator, GITAM University: led 10+ entrepreneurship events and speaker workshops, engaging 200+ students.